

Documentation file “_u_init.dat”

This document contains the documentation for the chemical data output file ending in “_u_init.dat” produced by the code REMIX. The file is written by the Chemistry class through the subroutine `write_conc_comp_tar_wat`. It reports the initial aqueous chemistry of the domain in terms of aqueous components and variable activity species, for both the domain (target) waters and the external (boundary/recharge) waters.

We describe the output file with the same `calcite_eq` style example used for the input files: a system with components built from the primary species, a domain of target waters and a single external inflow water named “entrada”. A representative file can be found alongside the simulation output.

Output file: “_u_init.dat”

This file contains the initial aqueous component and species concentrations of every water in the domain and of the external waters. It is written by the Chemistry class. The numeric values are written in scientific notation (format `ES13.5E2`); in every concentration block rows are components or species and columns are waters.

- **Block 1:** Number of aqueous components
 - `num_comps` (integer)
 - Number of aqueous components, taken from the reference (domain/initial/resident) water type
- **Block 2:** Aqueous components
 - **Line 2a** (repeated `num_comps` times): `uI = combination`
 - Each component written as the linear combination of variable activity species defined by row *I* of the component matrix `comp_mat_aq`
- **Block 3:** Aqueous variable activity species
 - **Line 3a** (repeated once per species): `j: sp_name`
 - *j*: index of the species
 - `sp_name`: name of the variable activity species (same order used in the concentration rows)
- **Block 4:** Aqueous component concentrations in the domain (rows = components, cols = target waters)
 - **Line 4a** (one row per component): `u(comp, :)` (real)
 - Component concentrations of every target water, one column per target water
- **Block 5:** Aqueous variable activity species concentrations in the domain (rows = species, cols = target waters)

- **Line 5a** (one row per species): $c(\text{species}, :)$ (real)
- Species concentrations of every target water, one column per target water
- **Block 6:** Aqueous component concentrations of external waters (rows = components, cols = external waters)
 - **Line 6a:** *name* (string)
 - Name of each external water (column header), e.g. *entrada*
 - **Line 6b** (one row per component): $u(\text{comp}, :)$ (real)
 - Component concentrations of the external (boundary/recharge) waters
- **Block 7:** Aqueous variable activity species concentrations of external waters (rows = species, cols = external waters)
 - **Line 7a:** *name* (string)
 - Name of each external water (column header)
 - **Line 7b** (one row per species): $c(\text{species}, :)$ (real)
 - Species concentrations of the external waters
- **End of file**

Number of aqueous components: 1

Aqueous components:

u1 = h+ + hco3-

Aqueous variable activity species:

1: h+

2: ca+2

3: o2(aq)

4: hco3-

Aqueous component concentrations in the domain (rows = components, cols = target waters):

7.05840E-01 7.05840E-01 7.05840E-01 7.05840E-01 7.05840E-01 7.05840E-01 7.05840E-01 7.05840E-01 7.05840E-01 7.05840E-01

Aqueous variable activity species concentrations in the domain (rows = species, cols = target waters):

1.00000E-05 1.00000E-05 1.00000E-05 1.00000E-05 1.00000E-05 1.00000E-05 1.00000E-05 1.00000E-05 1.00000E-05 1.00000E-05
 1.00000E-03 1.00000E-03 1.00000E-03 1.00000E-03 1.00000E-03 1.00000E-03 1.00000E-03 1.00000E-03 1.00000E-03 1.00000E-03
 1.00000E-06 1.00000E-06 1.00000E-06 1.00000E-06 1.00000E-06 1.00000E-06 1.00000E-06 1.00000E-06 1.00000E-06 1.00000E-06
 7.05830E-01 7.05830E-01 7.05830E-01 7.05830E-01 7.05830E-01 7.05830E-01 7.05830E-01 7.05830E-01 7.05830E-01 7.05830E-01

Aqueous component concentrations of external waters (rows = components, cols = external waters):

boundary
 5.38611E-05

Aqueous variable activity species concentrations of external waters (rows = species, cols = external waters):

boundary
 1.00000E-07
 1.00000E-09
 2.52947E-04
 5.37611E-05